SIH Problem 26042: Technical Constraints Analysis

This document outlines the critical constraints for developing an offline, AI-powered Vernacular Pedagogy and Real-Time Translation Tool (Problem ID: 26042). It includes explicit requirements from the problem statement and implicit engineering realities for deploying on 2GB RAM Android tablets.

Explicit Constraints

- **Offline Functionality:** The entire system must operate 100% offline after an initial one-time content synchronization. No network calls or cloud fallbacks are permitted during classroom use.
- **Hardware Baseline:** The target deployment device is a budget Android tablet running **Android 9.0 (API level 28)** with **≤ 2 GB of total RAM**.
- **Target Languages:** Source is Standard Hindi (Foundational Literacy and Numeracy curriculum content). Targets are Ho, Mundari, and Santhali. The prototype milestone requires a fully working pipeline for at least **one** tribal language.
- **Latency Threshold:** Real-time voice-to-voice translation must achieve an end-to-end latency of **≤ 3.0 seconds** from the moment the teacher stops speaking to the playback of synthesized tribal speech.
- **Curricular Alignment:** All auto-generated content (lesson scripts, activity instructions, bilingual worksheets, assessment prompts, visual flashcards) must strictly map to the **NIPUN Bharat** learning outcomes framework for foundational grades (Grades 1–3).
- **Deliverables:** A functional Android APK, a clear demonstration video showing on-device execution, and an open GitHub repository.

Implicit Hardware & Engineering Constraints

- **Usable Memory Budget (≤ 500 MB RAM):** A device with 2 GB total RAM devotes 1.2 GB to 1.5 GB to the Android OS, system services, and background daemons. Your entire app process (UI + STT + NMT + TTS runtimes) must stay under **450–500 MB peak RSS (Resident Set Size)** to prevent the Android Low Memory Killer (LMK) from terminating it.
- **CPU-Only Inference:** Low-cost government-issue tablets typically use entry-level quad-core SoCs (e.g., MediaTek Helio A22, Unisoc SC9863A) with no usable neural processing unit (NPU). All inference must run on quantized CPU backends (e.g., ONNX

Runtime, TFLite, or GGML using INT8/INT4 precision).
- **Storage & Download Footprint:** Low-end tablets often have 16 GB to 32 GB eMMC storage with limited free space. The base APK plus offline model weights, voice fonts, and dictionary files should ideally stay under **1.2 GB–1.5 GB** total disk footprint.
- **Thermal Throttling:** Running continuous local inference on cheap hardware generates heat rapidly. Under thermal throttling, CPU clock speeds drop by up to 40%, meaning a model that runs in 2.5 seconds on a cold tablet can take 4.5 seconds after 15 minutes of classroom use. Models must have low CPU utilization headroom.
- **Local PDF Generation:** Generating bilingual worksheets must occur strictly on-device without cloud rendering engines. The app must assemble layouts, embed fonts, and compile print-ready PDFs natively (e.g., via Android's PdfDocument API).

Implicit Latency Budget (The 3-Second Cascade)

To maintain ≤ 3.0 seconds total latency on an entry-level CPU, each step of the pipeline must adhere to strict time windows:

Pipeline Stage	Max Allocated Time	Engineering Reality
Voice Activity Detection (VAD) & Silence Padding	0.30s	Must quickly detect the end of speech without waiting for long pauses.
Speech-to-Text (STT - Hindi)	0.90s	Acoustic model must process short phrases (3–5 seconds audio) faster than real-time (RTF < 0.3).
Machine Translation (NMT - Hindi to Tribal)	0.60s	Constrained sequence length (< 25 tokens). Model must execute in single-pass greedy decoding.

Pipeline Stage	Max Allocated Time	Engineering Reality
Text-to-Speech (TTS - Tribal)	0.90s	Cannot use heavy diffusion or autoregressive vocoders; requires lightweight acoustic synthesis (e.g., Piper/FastSpeech2 + MelGAN/VITS quantized).
Audio I/O & Inter-Process Overhead	0.30s	Memory transfers, buffer allocations, and UI thread updates.
Total Target	**≤ 3.00s**	Zero safety margin; requires parallelized or streaming inference where feasible.

Implicit Linguistic & Pedagogical Constraints

- **Bundled Custom Unicode Fonts:** Android 9 does not natively ship system glyphs for tribal scripts such as **Ol Chiki** (for Santhali) or **Warang Citi** (for Ho). The app must package and render its own .ttf/.otf font assets to prevent missing-character rectangles ("tofu").
- **Dual-Script Flexibility:** While Santhali has the official Ol Chiki script, many regional schools and teachers in Jharkhand read tribal languages phonetically transliterated into Devanagari or Latin. The UI and generated worksheets must support toggling between native tribal script and Devanagari transliteration.
- **Extreme Low-Resource NLP:** There is practically zero web-scale parallel data for Mundari and Ho. A general-purpose translator will produce gibberish. The vocabulary must be restricted and heavily fine-tuned on the finite domain of primary education:

numbers, shapes, basic verbs, animals, everyday classroom instructions, and basic emotions.

- **Noisy Classroom Acoustic Environment:** Rural single-room or multi-grade schools have high ambient noise (reverberation, outdoor sounds, chatter from 30+ students). Standard open-mic setups will fail; the interaction model requires Push-to-Talk (PTT) with integrated digital noise gating.
- **Zero-Jargon Teacher UX:** The teacher is not an NLP engineer. The interface cannot require prompting, parameter tuning, or complex navigation. Key actions—translating a command, showing a flashcard, generating an exercise—must be executable in one or two taps.